DS-A2: Inference and Uncertainty in Black-tailed Godwit Migration Timing

1. Research question and analysis protocol

This analysis asks: **What is the difference in mean timing of southward crossing of 52°N between tracked juvenile and adult Black-tailed Godwits, and how uncertain is this estimate?**

The analysis continues DS-A1 using the public Dryad dataset associated with Verhoeven et al. The analytical subset comprises southward crossings of 52°N in 2016–2017. The outcome is the numerical day-of-year supplied in the `Date` column.

The estimand is the juvenile-minus-adult difference in **record-weighted mean crossing date**, expressed in days. Each migration record receives equal weight. Repeated records from the same bird are retained, with their dependence addressed during inference.

The hypotheses are:

- **H0:** There is no difference in mean crossing date between two age groups.
- **H1:** The mean crossing date is different.

A seven-day difference was selected as an interpretable educational threshold for a meaningful effect. This is a planning choice, not a validated ecological cutoff. Statistical significance and practical importance are distinguished.

The dataset and general age-group pattern had already been inspected during DS-A1. This analysis is therefore retrospective and is not presented as preregistered.

2. Data and reproducibility

The subset contains **78 migration records from 71 birds**. Seven adult birds contributed records from both years.

Age group	Migration records	Individual birds	Mean day-of-year	Record-level SD
Adult	50	43	176.96	10.50
Juvenile	28	28	225.32	13.04

The required fields contained no missing values. Bird-year combinations were unique, crossing dates fell within the valid day-of-year range, and each bird belonged to one age group within the subset. No imputation or outlier removal was performed.

The exact input file was identified by SHA-256:

`ef3b0967eac749481aaf7f9220ae617ff70daed3d8f63296dc326ca34947df59`

Random seeds were fixed separately for bootstrap, sampling-distribution, power and counterexample calculations. The notebook records dependency versions and saves numerical results and figures.

3. Inference on the observed difference

Cluster-bootstrap interval

Birds were resampled with replacement separately within each age group. Every selected bird contributed all its migration records. Selecting a bird multiple times repeated its complete record set.

For each replicate, the mean was calculated across all resampled records in each group. This preserved the record-weighted estimand rather than replacing it with an equally weighted mean of bird averages.

The procedure generated **10,000 bootstrap estimates**. Their 2.5th and 97.5th percentiles formed an approximate 95% confidence interval.

The observed juvenile-minus-adult difference was **48.36 days**. The bootstrap standard error was **2.89 days**, and the 95% interval was **42.72–54.08 days**.

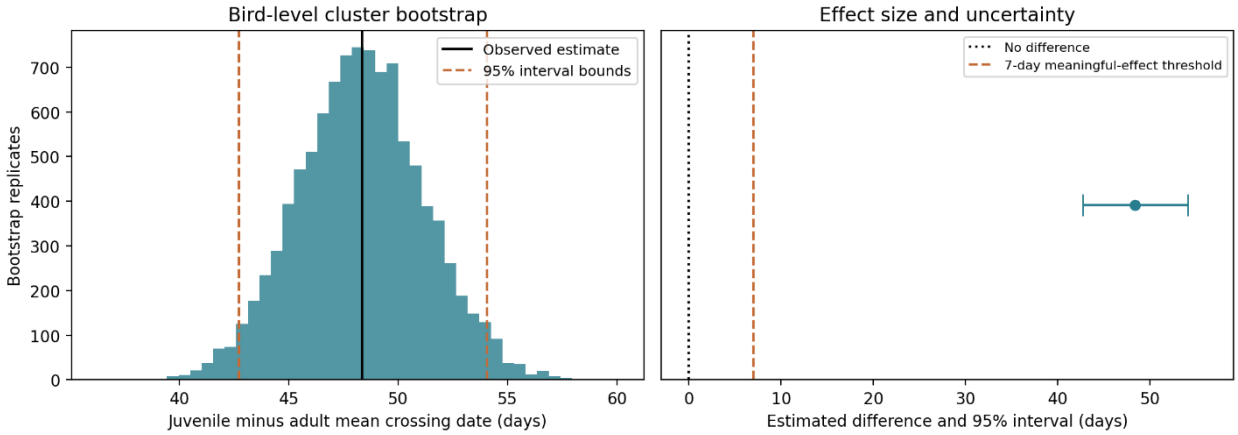


Figure 1. Bird-level cluster-bootstrap distribution and the estimated effect with its 95% percentile interval.

Observed-data hypothesis test

An additional two-sided test evaluated the zero-difference hypothesis using a cluster-robust standard error, with individual bird as the cluster.

Residuals were calculated relative to each age-group mean and summed within birds. The variance estimate used these cluster-level residual sums and the CR1 finite-sample correction:

\[\frac{G}{G-1}\frac{N-1}{N-2}, \]

where \(G=71\) birds and \(N=78\) records. The test statistic was the observed difference divided by its cluster-robust standard error. Inference used a t-distribution with \(G-1=70\) degrees of freedom.

The cluster-robust standard error was **2.92 days**, giving **t(70) = 16.56** and **p ≈ 6.47 × 10⁻²⁶**.

Both methods support a nonzero measured contrast under their assumptions. The bootstrap interval also lies entirely above the selected seven-day threshold. Neither method corrects selection or systematic measurement bias.

4. Sampling distribution under a known model

A separate simulation demonstrated how the estimated difference varies across repeated samples when the true effect is known.

A total of **3,000 studies** were generated, each containing 28 independent juvenile birds and 43 independent adult birds, with one observation per bird. Dates followed normal distributions with a true mean difference of **seven days**.

Standard deviations were calibrated from the observed data after averaging repeated dates within each bird: approximately **13.04 days for juveniles** and **10.45 days for adults**. This calibration did not change the primary observed-data analysis.

Under the simulated independent-group model, the theoretical standard error is:

\[SE=\sqrt{\frac{\sigma_J^2}{n_J}+\frac{\sigma_A^2}{n_A}}. \]

The mean simulated estimate was **7.06 days**. The empirical standard error was **2.95 days**, close to the theoretical value of **2.93 days**.

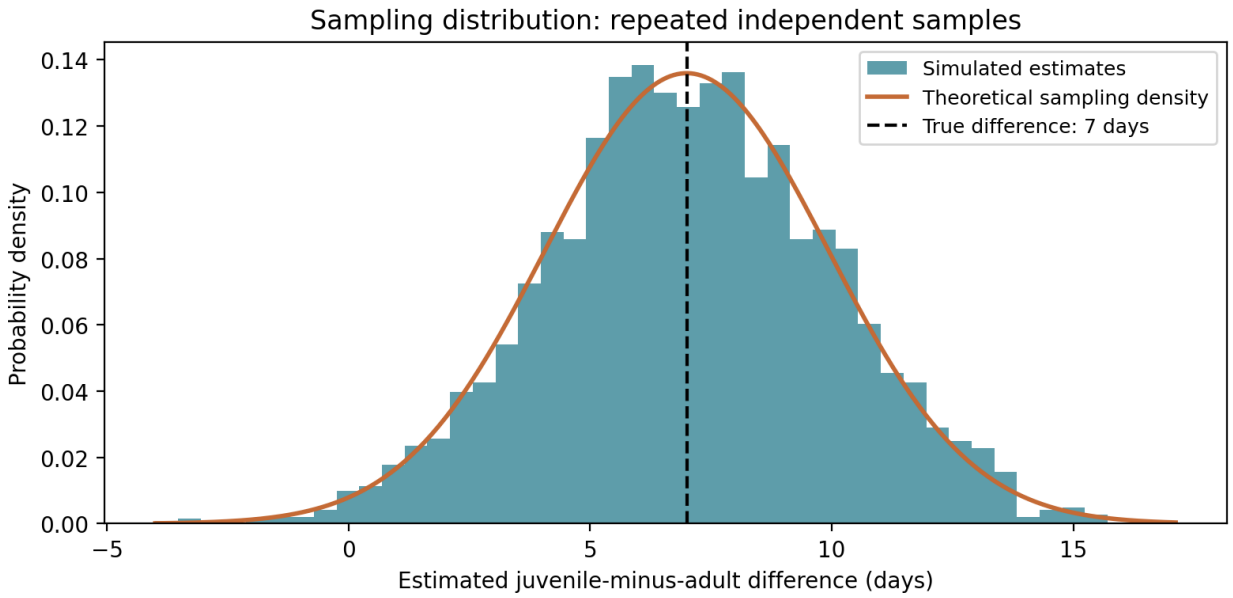


Figure 2. Sampling distribution of the estimated difference under a known seven-day effect, compared with its theoretical normal density.

The agreement supports the implementation under the specified model. It does not establish normality or independence in the original tracking data. The distribution describes variability between estimated mean differences, rather than between individual migration dates.

5. Power analysis

Power was evaluated for a prospective design with one observation per independent bird. Each scenario comprised 3,000 simulated studies analysed using a two-sided Welch t-test at **$\alpha = 0.05$**.

Three factors were varied:

- **Sample size:** 10, 20, 40 or 80 birds per age group.
- **Noise:** 0.5, 1 or 2 times the calibrated group standard deviations.
- **True difference:** 0, 3.5, 7 or 14 days.

For nonzero effects, power was estimated as the proportion of studies with $p < 0.05$. At zero effect, this proportion estimated the false-positive rate.

Birds per group	Power for a seven-day effect at baseline noise
10	22.7%
20	43.3%
40	74.3%
80	95.8%

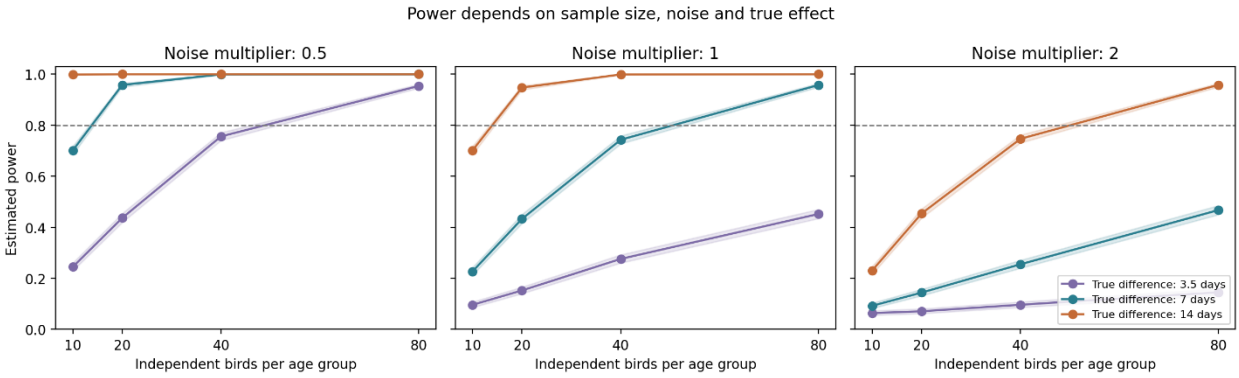


Figure 3. Estimated power across sample sizes, noise levels and true effects. Shaded bands show 95% Wilson intervals for Monte Carlo uncertainty.

Power increased with sample size and effect size and decreased with noise. A nonsignificant result in a small sample therefore does not establish absence of a meaningful difference.

Zero-effect rejection rates ranged from approximately **4.4% to 5.7%**; their Monte Carlo intervals included the nominal 5% level. These calculations are conditional planning scenarios, not estimates of the original clustered study's power or definitive fieldwork sample-size requirements.

6. Assumption-violating counterexample

The counterexample violated the assumption of independent observations.

In 3,000 simulated zero-effect studies, each group contained 20 independent birds. Each study was first analysed correctly using one observation per bird. Every observation was then copied five times, and the copies were incorrectly analysed as independent observations.

Copying increased the number of rows from 20 to 100 per group without adding birds or independent information.

Analysis	Birds per group	Rows per group	False-positive rate
Independent observations	20	20	4.7%
Copies treated as independent	20	100	37.6%

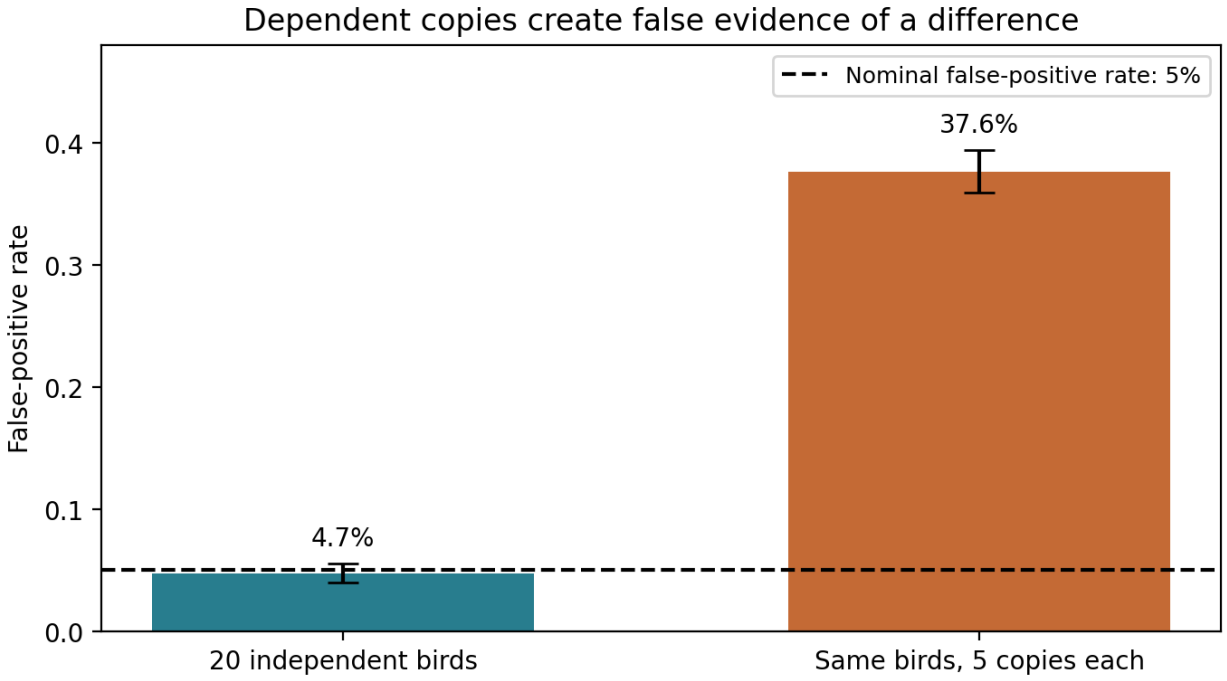


Figure 4. False-positive rates under a zero true effect, before and after dependent copies are treated as independent. Error bars show Monte Carlo intervals.

The changed conclusion is consequential: the naive analysis frequently produces apparently significant evidence when no true difference exists.

Exact copying represents perfect dependence and is an intentionally extreme stress test. It does not estimate error inflation in the original dataset. In the observed-data analysis, repeated records were handled by bird-level bootstrap resampling and clustered standard errors.

7. Verification and limitations

Numerical verification included direct recalculation of group-mean differences, comparison of a manually calculated Welch statistic and p-value with SciPy, and confirmation that the bootstrap reproduced identical results with the same seed. The simulated sampling standard error was also compared with its theoretical value.

Important limitations remain:

- **Selection:** Usable tracking records represent a selected subset of tagged birds. Mortality, recapture and device retrieval can influence inclusion.
- **Measurement:** Juvenile and adult timing was obtained using different tracking systems. Device-specific bias and individual-date reconstruction uncertainty were not corrected.

- **Years:** The effect is a pooled 2016–2017 contrast, not a year-adjusted age coefficient. Future-year variability is not quantified.
- **Assumptions:** Observed-data inference assumes independence between birds. Power simulations additionally assume normal distributions and one observation per independent bird.
- **Meaningful-effect threshold:** Seven days is an educational planning choice rather than an established ecological criterion.

The results support a difference in recorded timing within the analysed sample under the stated assumptions. They do not establish its biological cause, a causal effect of age, or universal applicability to other populations and years.

The counterexample illustrates why more rows need not mean more independent information. Likewise, a small p-value does not guarantee unbiased or representative evidence.

8. AI-use disclosure

ChatGPT assisted with Python codes.

Accepted recommendations included resampling complete bird histories, distinguishing real-data inference from prospective power simulations, and documenting the retrospective status of the analysis.

Independent-row inference was rejected as the primary real-data method because some birds contributed repeated records. It was used only in the explicitly labelled counterexample.

The verification code was also AI-assisted.

References

Verhoeven, M. A. et al. (2021). *Age-dependent timing and routes demonstrate developmental plasticity in a long-distance migratory bird* [Dataset]. Dryad. https://doi.org/10.5061/dryad.02v6wwq4j. Dataset version: 3 December 2021. CC0.

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